



FRACA SERVCOM INVENTORY MANAGEMENT SYSTEM

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End Year Project

Chapter 1: Introduction

1.1 Background

FRACA SERVCOM is a medium-sized business dealing in furniture (office and home) and hardware supplies such as door locks, ceiling lights, hinges, handles, and other fittings. The company operates two main divisions—Furniture and Hardware—and currently records inventory using manual or spreadsheet-based methods. This manual process has led to challenges such as inaccurate stock counts, delayed reordering, difficulty tracking sales, and time-consuming reporting. To address these inefficiencies, there is a need for an automated Inventory Management System that can simplify stock tracking, streamline purchasing and sales operations, and provide timely reports.

1.2 Problem Statement

The existing manual stock tracking method at FRACA SERVCOM is inefficient and error-prone. It is difficult to monitor real-time stock levels, identify low-stock items, and generate accurate reports. These limitations result in stockouts, overstocking, and delayed customer service.

1.3 Proposed Solution

The proposed solution is a web-based Inventory Management System developed using PHP, MySQL, HTML, CSS, and JavaScript. The system will centralize product data, automatically update stock after purchases and sales, generate low-stock alerts, provide sales and purchase reports, and restrict access based on user roles (Admin, Storekeeper, Cashier). This system will enhance stock accuracy, improve operational efficiency, and support faster decision-making.

1.4 Objectives

Main Objective: To design and develop an Inventory Management System for FRACA SERVCOM that automates stock tracking and reporting processes.

Specific Objectives:

1. To collect user requirements and analyze current inventory practices.
2. To design a database and user interface for managing products, purchases, and sales.
3. To develop modules for real-time stock monitoring and low-stock alerts.
4. To generate basic reports for inventory, sales, and purchases.
5. To test and validate the system with real data from FRACA SERVCOM.

1.5 Justification

The project is significant as it will reduce human error, save time, and improve accuracy in inventory management. For FRACA SERVCOM, it provides an affordable solution tailored to its specific business needs without depending on costly commercial software. For society, it demonstrates how small and medium enterprises can digitize operations affordably, contributing to economic growth and digital transformation. Academically, it offers a learning opportunity in web-based system design and development.

1.6 Scope

In-scope: Product management, stock monitoring with low-stock alerts, purchase and sales management, user login and role management, and basic reporting. Out of scope: Multi-branch management, mobile app or online shop integration, and advanced analytics or accounting features.

Chapter 2: Literature Review

Several existing systems, including Odoo, ERPNext, and Zoho Inventory, offer comprehensive inventory and warehouse management features (Odoo S.A., n.d.; ERPNext, n.d.; Zoho Corporation, n.d.). However, these solutions are complex and expensive, often requiring online access and technical expertise, which may not suit small or medium-sized businesses like FRACA SERVCOM. Lightweight systems such as Square for Retail and Loyverse POS focus on ease of use and retail inventory management (Square, Inc., n.d.; Loyverse POS, n.d.). While simple, they still depend on cloud subscriptions and are not easily customizable. The proposed system will differ by providing a localized, low-cost, and easy-to-use solution tailored to FRACA SERVCOM's needs. It builds upon principles from existing systems—such as low-stock alerts and sales tracking—while addressing gaps in affordability, offline functionality, and domain-specific customization for furniture and hardware products.

References (APA Style)

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Chapter 3: Methodology

3.1 Development Methodology

The project will adopt an Incremental Development Approach, which allows building and testing the system in stages. This method is ideal for a single-developer project within a limited timeframe, ensuring that essential features are delivered early and improved progressively.

3.2 Phases of Development

- 1) Requirements Analysis – Gather system needs through observation and interviews with FRACA SERVCOM staff.
- 2) System Design – Create ERDs, DFDs, and wireframes for the database and UI.
- 3) Implementation – Develop the frontend using HTML, CSS, and Bootstrap; backend using PHP or Laravel; database in MySQL.
- 4) Testing – Conduct unit, system, and user acceptance testing.
- 5) Deployment & Documentation – Install system locally and prepare manuals.

3.3 Justification for Methodology

The incremental method enables continuous feedback, supports flexibility for new features, and suits the project's short timeline and single-developer setup.

Appendices

A. Required Resources

- Computer – 4GB RAM, Core i3+, 500GB HDD – System development
- Software – XAMPP, VS Code, Bootstrap – Development tools
- Database – MySQL – Data storage
- Internet Connection – 10 Mbps – Research and testing

B. Estimated Budget

Internet Access	Kshs. 1,000
Flash Disk	Kshs. 500
Stationery/Printing	Kshs. 500
Total Estimated Cost	Kshs. 2, 500

C. Project Schedule

Proposal Development	Oct 12 th – Nov 1st
System Analysis & SRS	
System Design	
System Development	
System Testing	
System Presentation & Submission of Project Documentation	

D. Project Risks and Mitigation

- System crash – Data loss – Regular backups
- Power outage – Delayed work – Use UPS/power backup
- Laptop theft – Loss of code – Use GitHub cloud backup